

Math 10A

Quiz 11; Tuesday, 8/7/2018

Time: 3 PM

Instructor: Roy Zhao

Name: _____

Circle True or False. (1 point for correct answer, 0 if incorrect)

1. **TRUE** False If 1 is an eigenvalue of A , then there is a nonzero vector $\vec{v}$ such that $A\vec{v} = \vec{v}$.

Solution: If 1 is an eigenvalue, then there is a vector such that $A\vec{v} = \lambda\vec{v} = 1\vec{v} = \vec{v}$.

2. True **FALSE** If two matrices A, B have the same eigenvalues, then they have the same solutions to $\vec{y}' = A\vec{y}$.

Solution: The two matrices could have the same eigenvalues but different eigenvectors.

Show your work and justify your answers. Please circle or box your final answer.

3. (10 points) (a) (8 points) Solve the system given by $\begin{cases} y_1'(t) = y_1(t) - 2y_2(t) \\ y_2'(t) = -4y_1(t) - y_2(t) \end{cases}$ and $y_1(0) = 3, y_2(0) = 0$.

Solution: We write this as $\vec{y}' = A\vec{y}$ with $\vec{y} = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$ and $A = \begin{pmatrix} 1 & -2 \\ -4 & -1 \end{pmatrix}$. Then we find the eigenvalues of A which are solutions to $(1 - \lambda)(-1 - \lambda) - (-2)(-4) = \lambda^2 - 1 - 8 = \lambda^2 - 9 = 0$. The solutions are $\lambda = \pm 3$. The eigenvector for $\lambda = 3$ is $\vec{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and the eigenvector for $\lambda = -3$ is $\vec{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Therefore, the general solution is

$$\vec{y} = c_1 e^{3t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -c_1 e^{3t} + c_2 e^{-3t} \\ c_1 e^{3t} + 2c_2 e^{-3t} \end{pmatrix}.$$

Now plugging in our initial conditions, we have that $y_1(0) = 3$ or $-c_1 e^0 + c_2 e^0 = -c_1 + c_2 = 3$ and $y_2(0) = 0$ so $c_1 e^0 + 2c_2 e^0 = c_1 + 2c_2 = 0$. Adding them gives $3c_2 = 3$ or $c_2 = 1$ and hence $c_1 = c_2 - 3 = -2$. Therefore, our solution is

$$\begin{cases} y_1(t) = 2e^{3t} + e^{-3t} \\ y_2(t) = -2e^{3t} + 2e^{-3t} \end{cases}.$$

- (b) (2 points) Find the least square error of using $y = x + 1$ as the line of best fit through the points $(0, 0)$, $(1, 2)$, $(2, 3)$.

Solution: To find the least square error, we sum up $(\text{expected} - \text{actual})^2 = ((0 + 1) - 0)^2 + ((1 + 1) - 2)^2 + ((2 + 1) - 3)^2 = 1^2 + 0^2 + 0^2 = 1$.